

CSE4227 Digital Image Processing

Lecture 02 – Digital image Fundamentals

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Today's Contents

□ Image Sensing and Acquisition

- Visual Perception
- Image formation model

• Chapter 2 from R.C. Gonzalez and R.E. Woods, Digital Image Processing (3rd Edition), Prentice Hall, 2008 [**Section 2.1, 2.2, 2.3**]

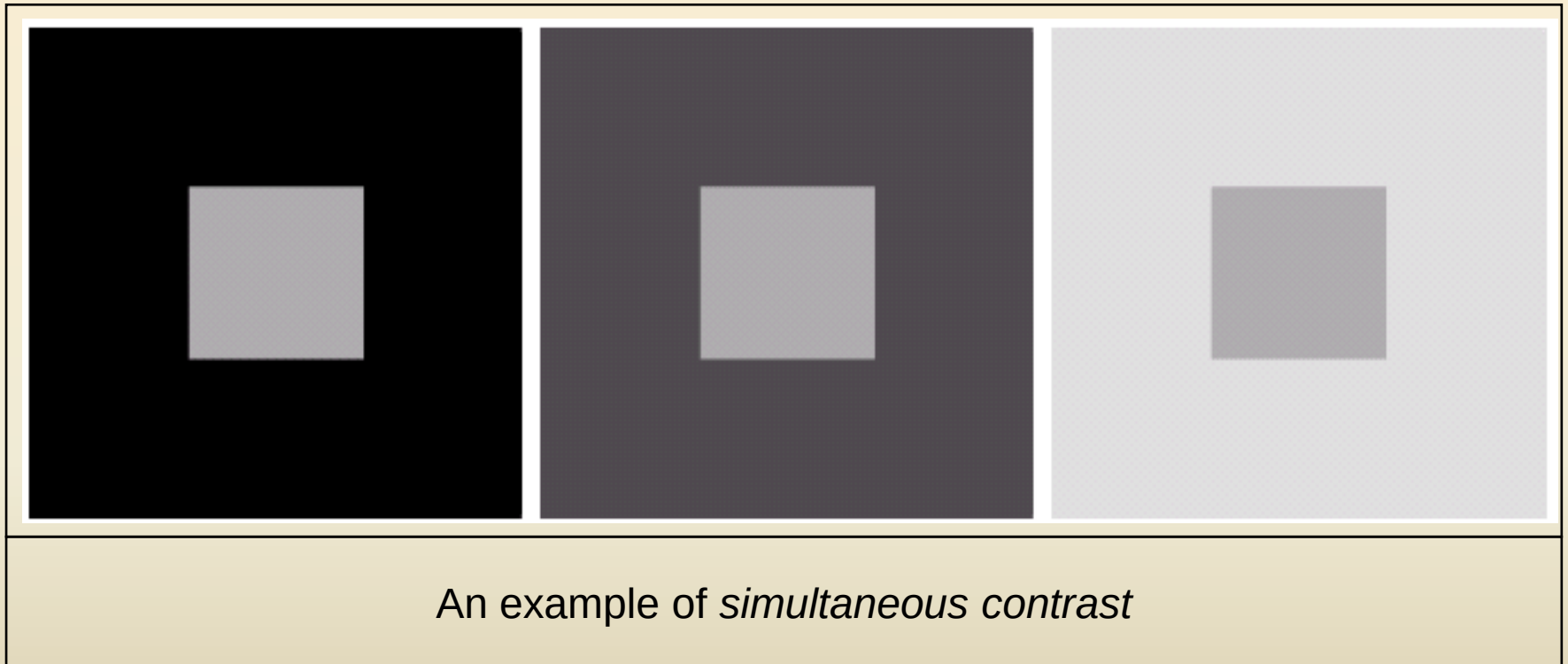
What is meant by visual perception?

- ❑ **Visual perception** is the ability to **see** and **interpret** (analyze and give meaning to) the **visual** information that surrounds us.
- ❑ There is a large difference between the image we **display** and the image we **actually perceive**.

Human Visual System

- ❑ There is a difference between the **luminance** of a pixel on a computer screen and the **perceived brightness** of this pixel.
- ❑ If we **double the screen luminance**, this **does not imply** that the perceived brightness is doubled too.
- ❑ The brightness also depends on other factors, such as **contrast around the pixel** and various **cognitive processes**

Brightness Adaptation & Discrimination (cont...)



Human Visual System

- ❑ The human visual system can perform a number of image processing tasks in a manner vastly superior to anything we are presently able to do with computers.
- ❑ The best vision model we have!
- ❑ If we want to mimic such processing, we need to carefully study the way our eyes and brain do this.

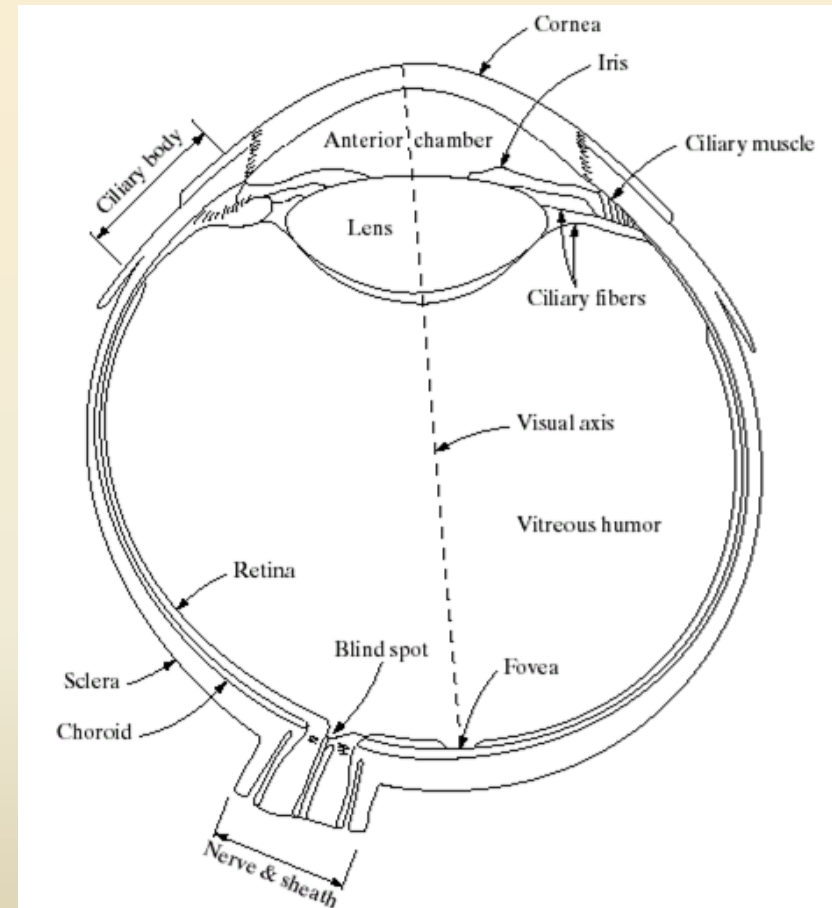
Human Visual System

- ❑ The human visual system consists of two functional parts:
 - **eye and**
 - **(part of the) brain.**

- ❑ The brain does all of the complex image processing, while the eye functions as the biological equivalent of a camera.

Structure Of The Human Eye

- ❑ The lens focuses light from objects onto the retina
- ❑ The retina is covered with light receptors called *cones* (6-7 million) and *rods* (75-150 million)
- ❑ Cones are concentrated around the fovea and are very sensitive to colour
- ❑ Rods are more spread out and are sensitive to low levels of illumination



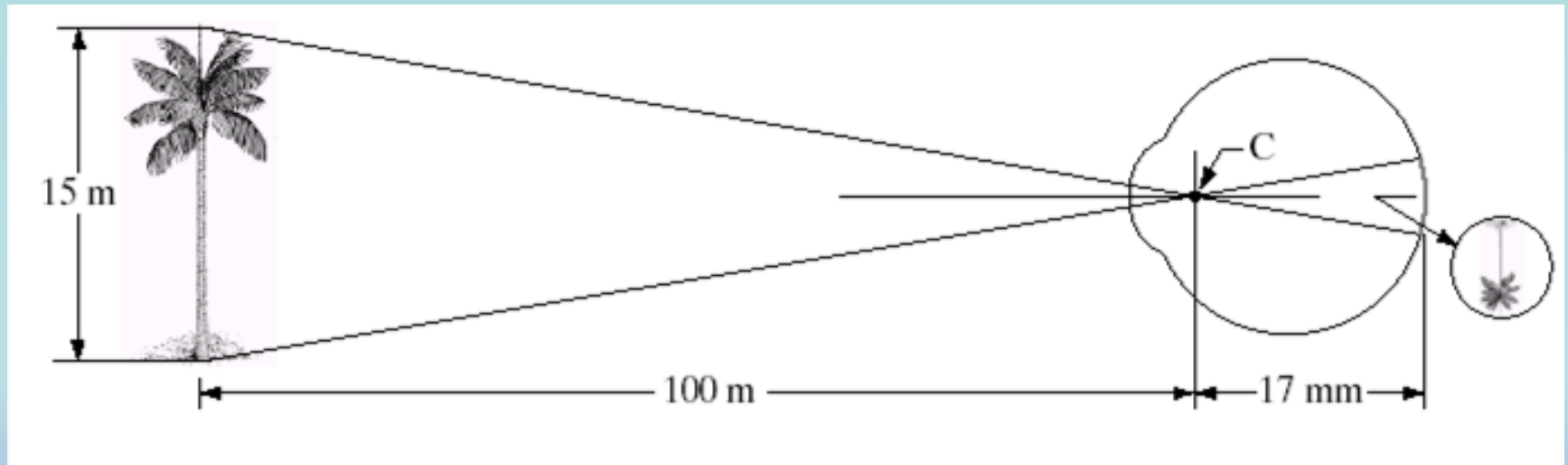
- When a light ray hits the eye, it will **first pass through the cornea, then** subsequently through the aqueous humor, the iris, the lens, and the vitreous humor before **finally reaching the retina.**
- The **cornea** is a transparent protective layer, which acts as a lens and **refracts the light.**
- The **iris** forms a round aperture that can vary in size and so **determines the amount of light** that can pass through.
- In the **retina**, the light rays are detected and **converted to electrical signals** by photoreceptors.

Visual Perception: Human Eye

- ❑ The **lens** contains 60-70% water, 6% of fat.
- ❑ The **iris** diaphragm controls amount of light that enters the eye.
- ❑ **Light receptors** in the **retina are Cones and Rods:**
 - **Cones** for bright light vision called **photopic**
 - Cones involve in color vision.
 - Cones are concentrated in **fovea** about 1.5x1.5 mm².
 - **Rods** for dim light vision called **scotopic**
 - Rods are sensitive to low level of light and
 - are not involved in color vision.
- ❑ **Blind spot** is the region of emergence of the optic nerve from the eye. In this area receptors are absent and no image detection.

Image Formation In The Eye

- ❑ Muscles within the eye can be used to change the shape of the lens allowing us focus on objects that are near or far away
- ❑ An image is focused onto the retina causing rods and cones to become excited which ultimately send signals to the brain



$$\text{(Height of the tree in the retinal image) / 17} = (15/100)$$

Some psychophysics:

Blind-Spot Experiment

☐ Draw an image similar to that below on a piece of paper (the dot and cross are about 6 inches apart)



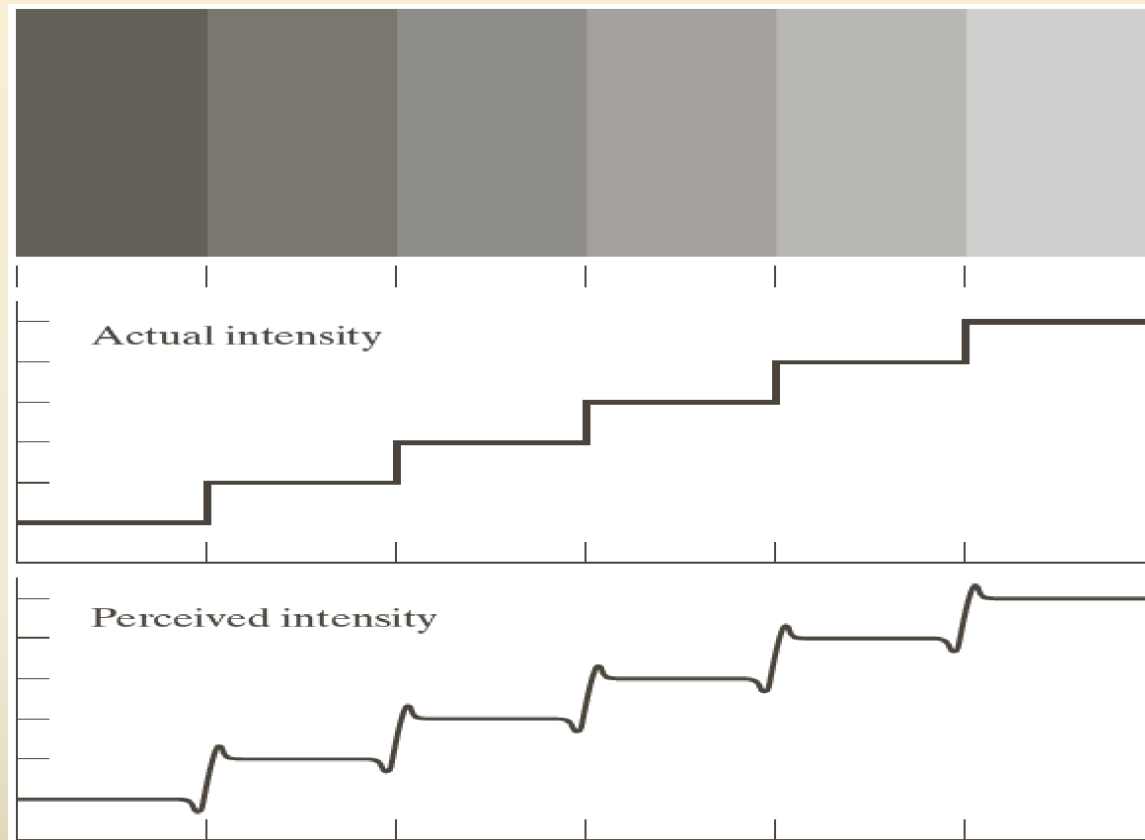
- ☐ Close your right eye and focus on the cross with your left eye
- ☐ Hold the image about 20 inches away from your face and move it slowly towards you
- ☐ The dot should disappear!

Brightness Adaptation & Discrimination

- ❑ The human visual system can perceive approximately 10^{10} different light intensity levels
- ❑ However, at any one time we can only discriminate between a much smaller number – *brightness adaptation*
- ❑ Similarly, the *perceived intensity* of a region is related to the light intensities of the regions surrounding it

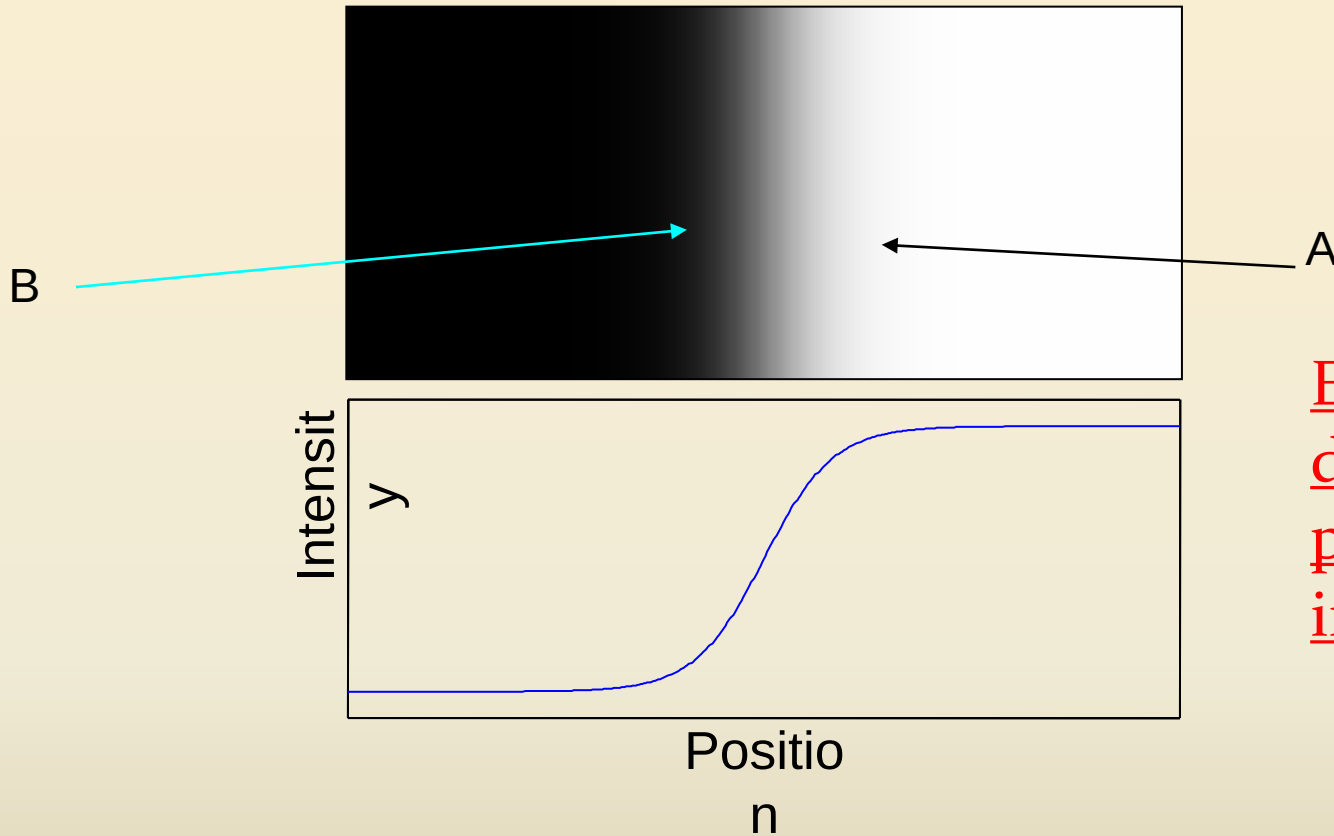
Brightness Adaptation & Discrimination

Intensities of surrounding points effect perceived brightness at each point.



In this image, edges between bars appear brighter on the right side and darker on the left side.

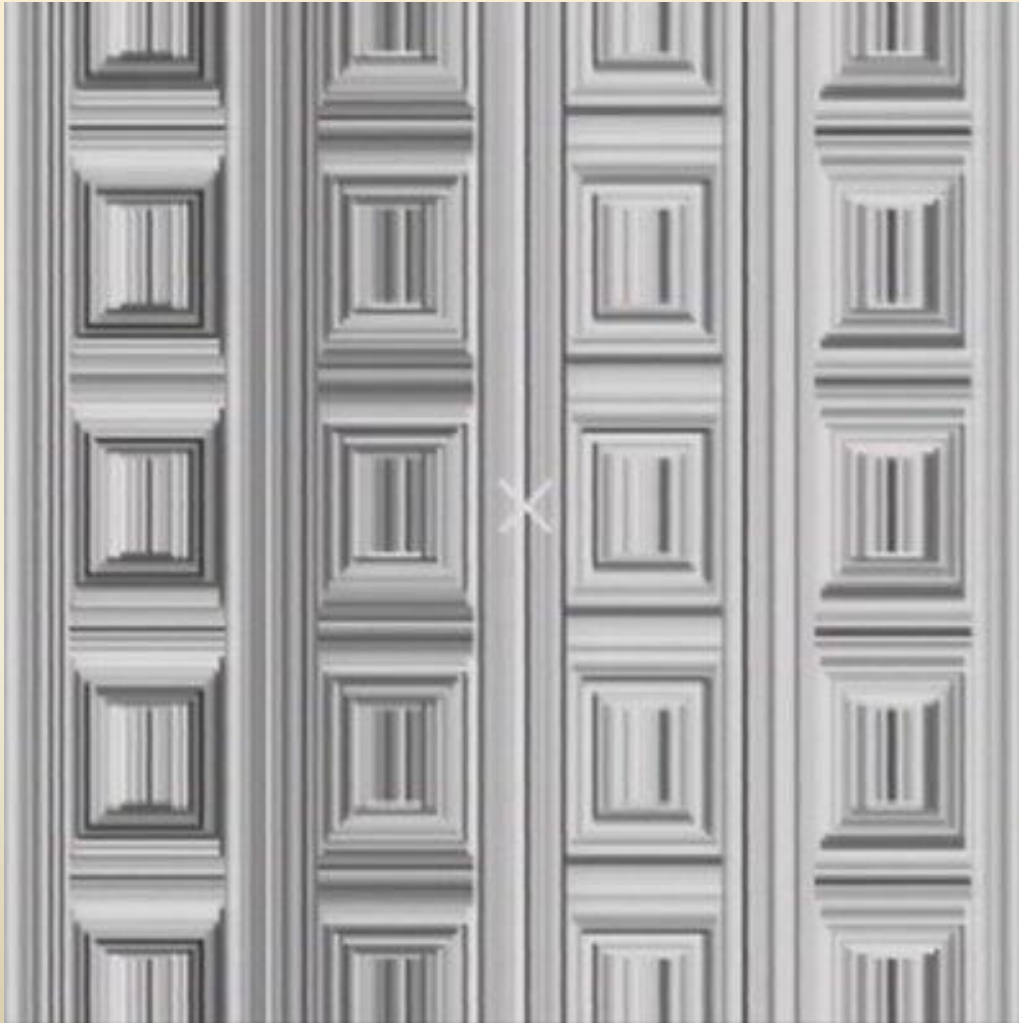
Mach Band Effect (Cont)



Ernst Mach
described this
phenomenon
in 1865

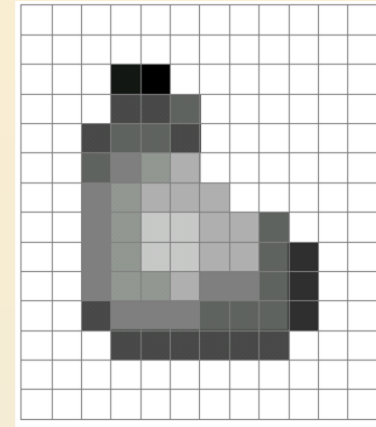
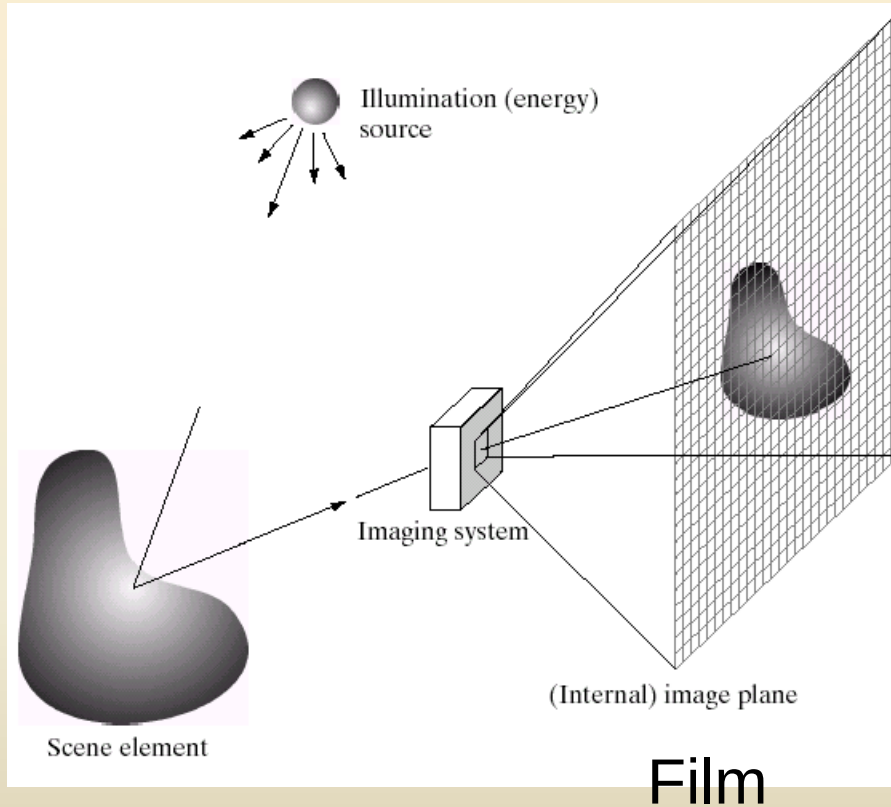
- ❑ In area A, brightness perceived is darker while in area B is brighter.
- ❑ This phenomenon is called ***Mach Band Effect***.

Optical Illusions

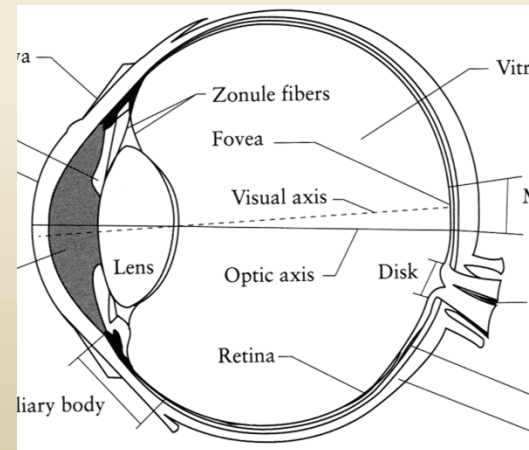


Stare at the cross in the middle of the image and think circles.

Image Formation



Digital Camera



The Eye

IMAGE SENSING AND ACQUISITION

In the following slides we will consider what is involved in capturing a digital image of a real-world scene

- ❑ Image sensing and representation and acquisition
- ❑ Sampling and quantisation (done in previous slide)
- ❑ Resolution

Image Acquisition

□ Three main elements

1. **Illumination source**
2. **Scene**
3. **Sensor (imaging system)**

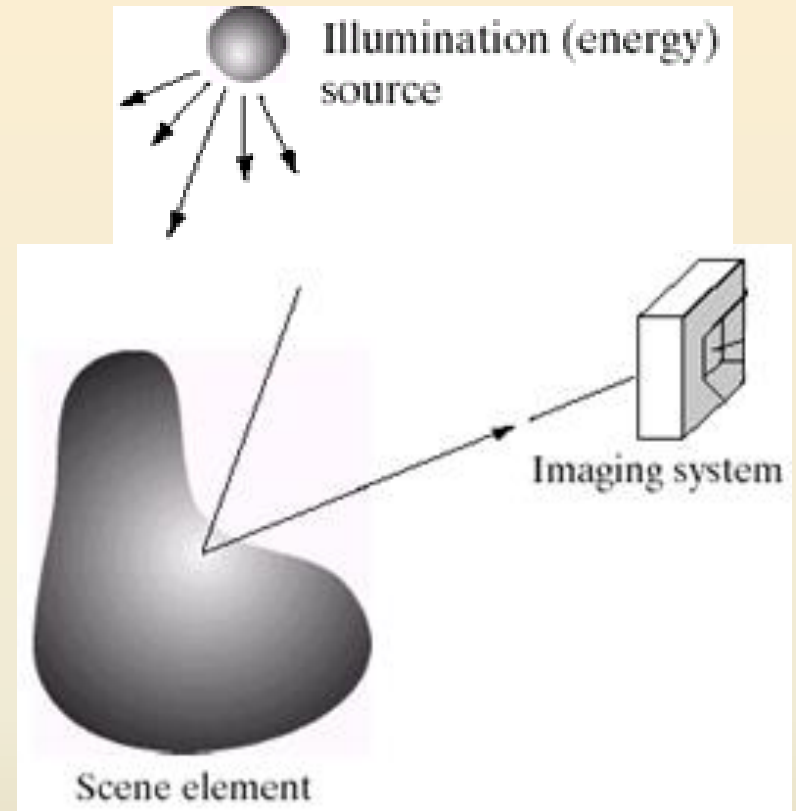
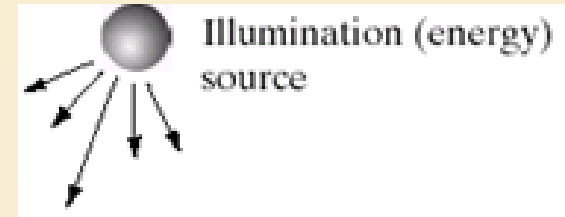


Image Acquisition

1. Illumination source:

- Can be light energy or
- EM spectrum
- Even less tradition sources like
 - Sound, heat

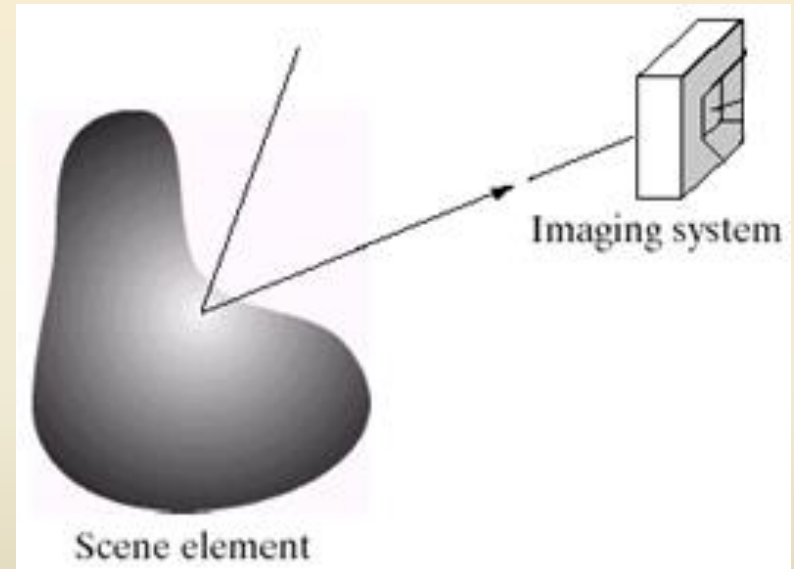


2. Scene:

- Any object: visible or hidden
- Source itself

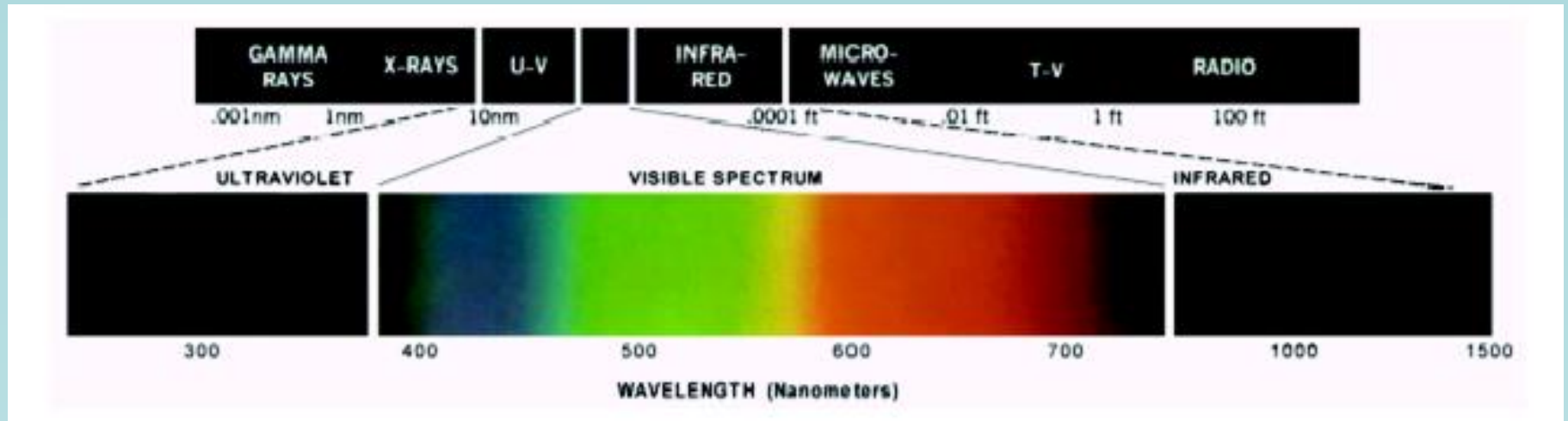
3. Sensor:

- Should be capable of sensing the energy



Light And The Electromagnetic Spectrum

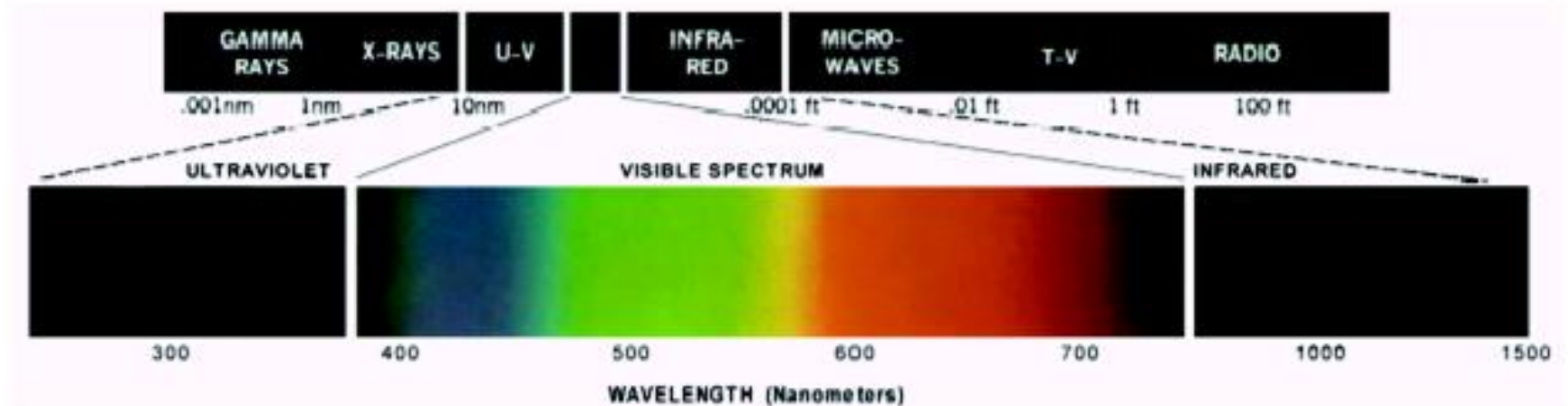
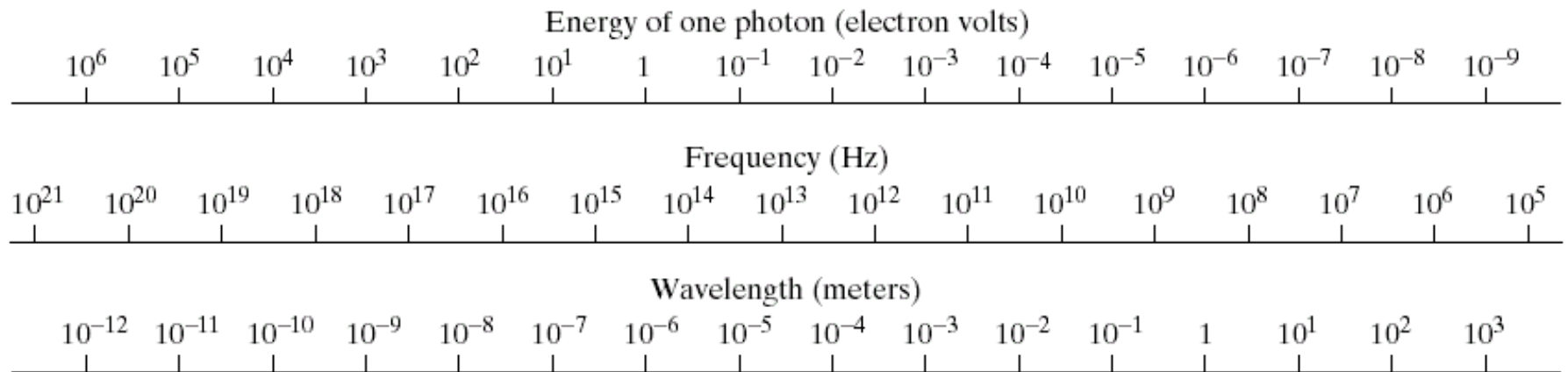
❑ In 1666 Sir Isaac Newton discovered that sunlight passed through a prism splits into a continuous spectrum of colors



❑ Light is just a particular part of the electromagnetic spectrum that can be sensed by the human eye

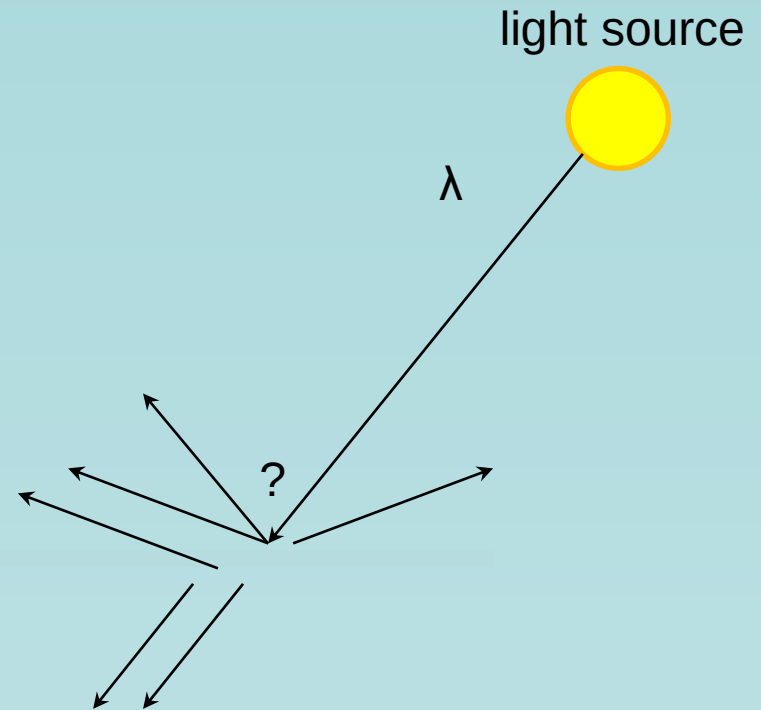
❑ The electromagnetic spectrum is split up according to the wavelengths of different forms of energy

- ❑ A discrete bundle (or *quantum*) of electromagnetic (or light) energy, **PHOTON** is proportional to frequency.



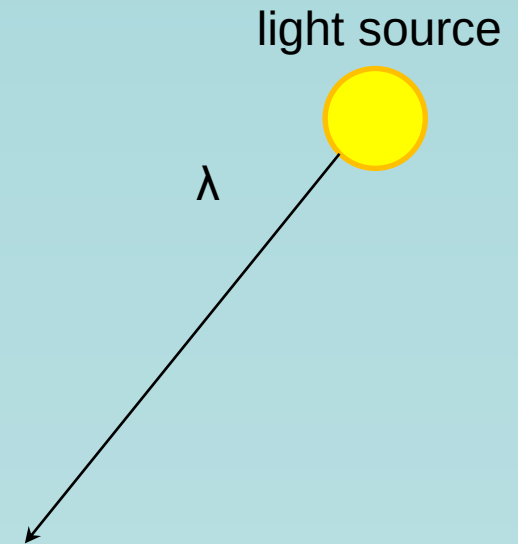
A photon's life choices

- ☐ Absorption
- ☐ Diffusion
- ☐ Reflection
- ☐ Transparency
- ☐ Refraction
- ☐ Fluorescence
- ☐ Subsurface scattering
- ☐ Phosphorescence
- ☐ Interreflection



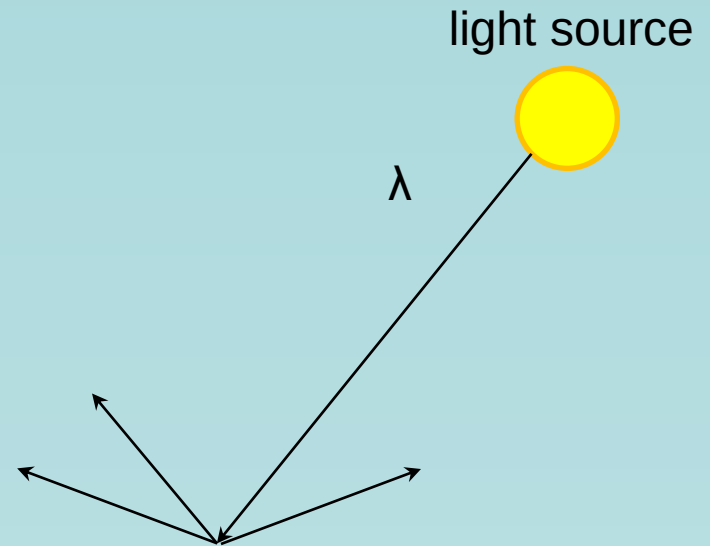
A photon's life choices

- **Absorption**
- Diffusion
- Reflection
- Transparency
- Refraction
- Fluorescence
- Subsurface scattering
- Phosphorescence
- Interreflection



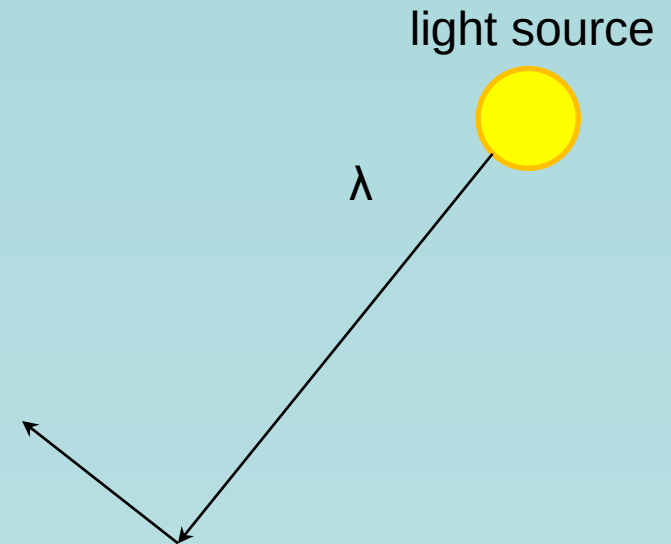
A photon's life choices

- Absorption
- **Diffuse Reflection**
- Reflection
- Transparency
- Refraction
- Fluorescence
- Subsurface scattering
- Phosphorescence
- Interreflection



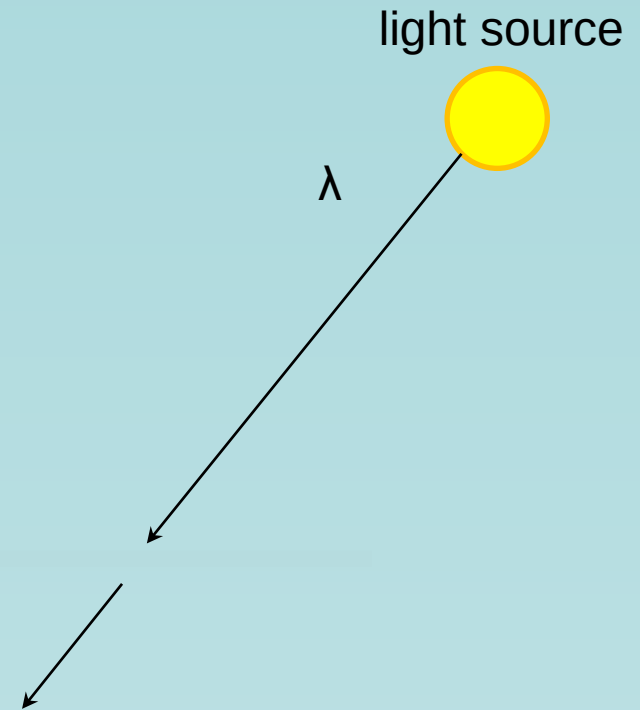
A photon's life choices

- Absorption
- Diffusion
- **Specular Reflection**
- Transparency
- Refraction
- Fluorescence
- Subsurface scattering
- Phosphorescence
- Interreflection



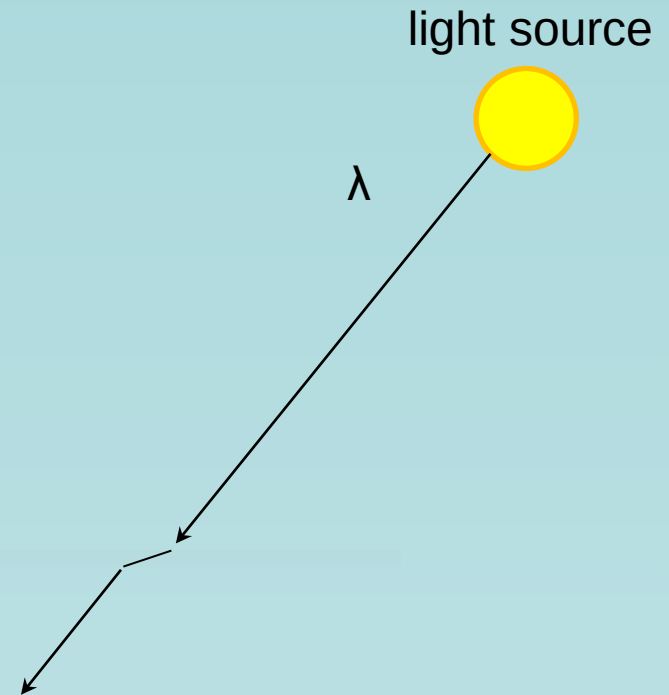
A photon's life choices

- Absorption
- Diffusion
- Reflection
- **Transparency**
- Refraction
- Fluorescence
- Subsurface scattering
- Phosphorescence
- Interreflection



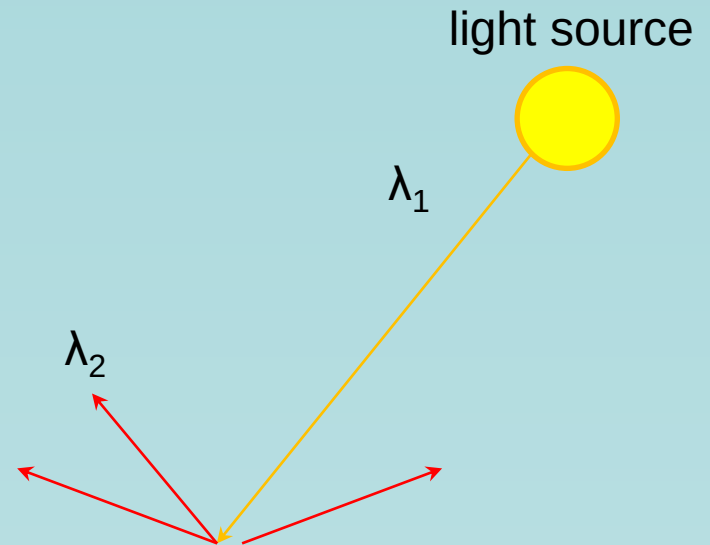
A photon's life choices

- Absorption
- Diffusion
- Reflection
- Transparency
- **Refraction**
- Fluorescence
- Subsurface scattering
- Phosphorescence
- Interreflection



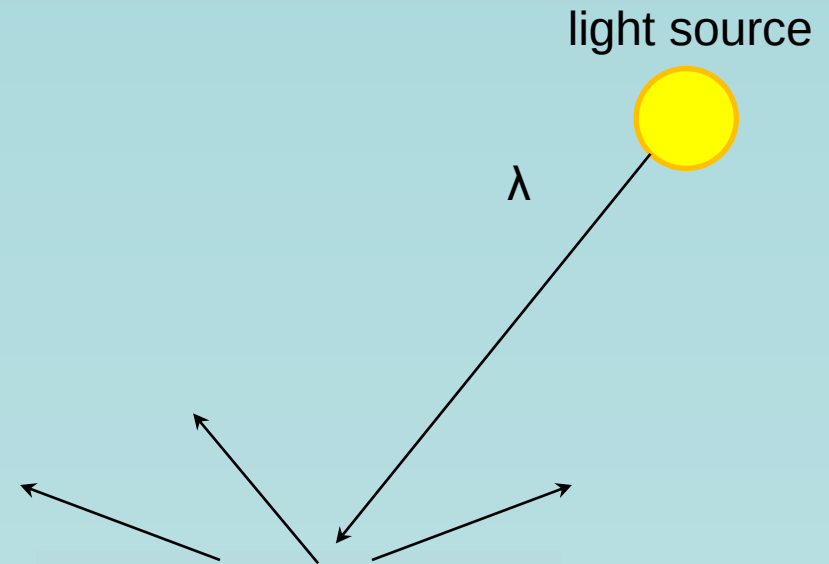
A photon's life choices

- Absorption
- Diffusion
- Reflection
- Transparency
- Refraction
- **Fluorescence**
- Subsurface scattering
- Phosphorescence
- Interreflection



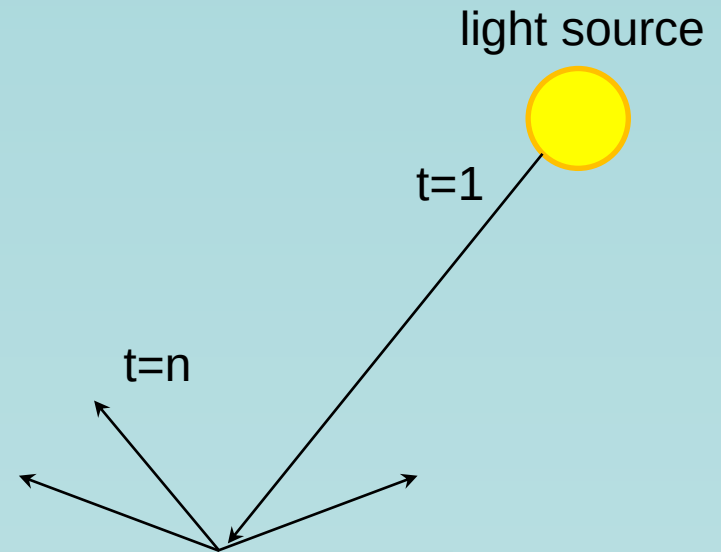
A photon's life choices

- Absorption
- Diffusion
- Reflection
- Transparency
- Refraction
- Fluorescence
- **Subsurface scattering**
- Phosphorescence
- Interreflection



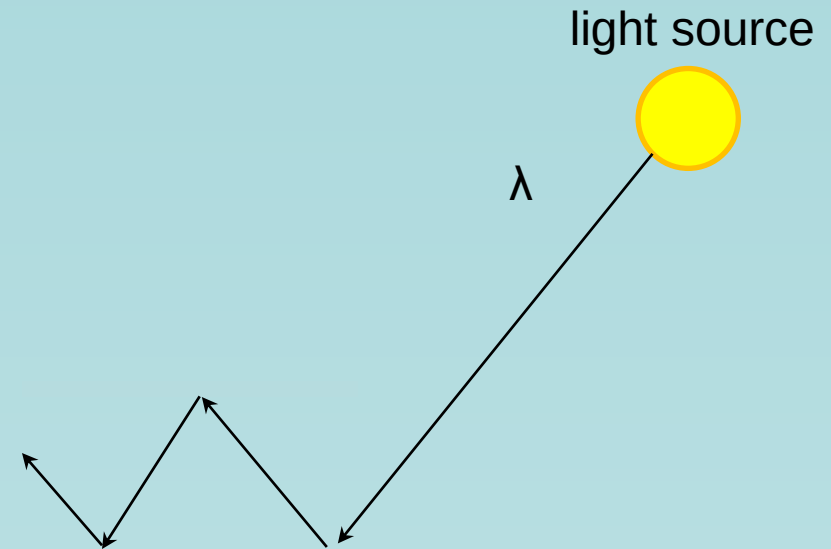
A photon's life choices

- Absorption
- Diffusion
- Reflection
- Transparency
- Refraction
- Fluorescence
- Subsurface scattering
- **Phosphorescence**
- Interreflection



A photon's life choices

- Absorption
- Diffusion
- Reflection
- Transparency
- Refraction
- Fluorescence
- Subsurface scattering
- Phosphorescence
- **Interreflection**



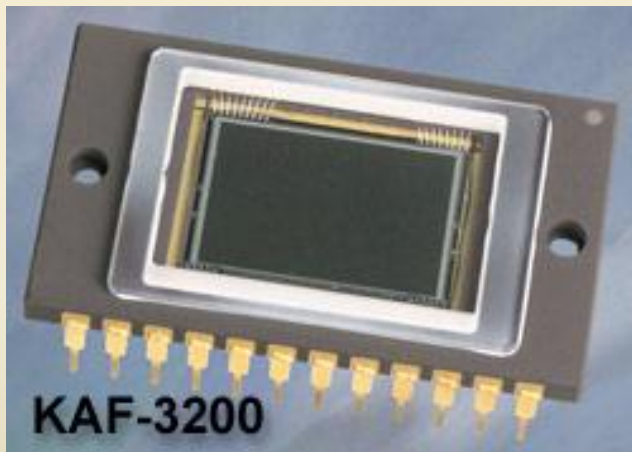
(Specular Interreflection)

Image Sensors

- ❑ Incoming energy lands on a sensor material responsive to that type of energy and this generates a voltage
- ❑ Collections of sensors are arranged to capture images

Charge-Coupled Device (CCD)

- ✓ Used for convert a continuous image into a digital image
- ✓ Contains an array of light sensors
- ✓ Converts photon into electric charges accumulated in each sensor unit

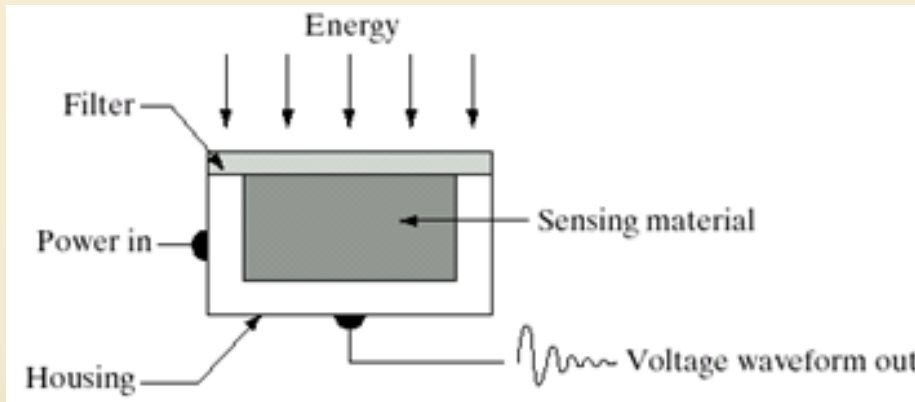


CCD KAF-3200E from
Kodak.
(2184 x 1472 pixels,
Pixel size 6.8 microns²)

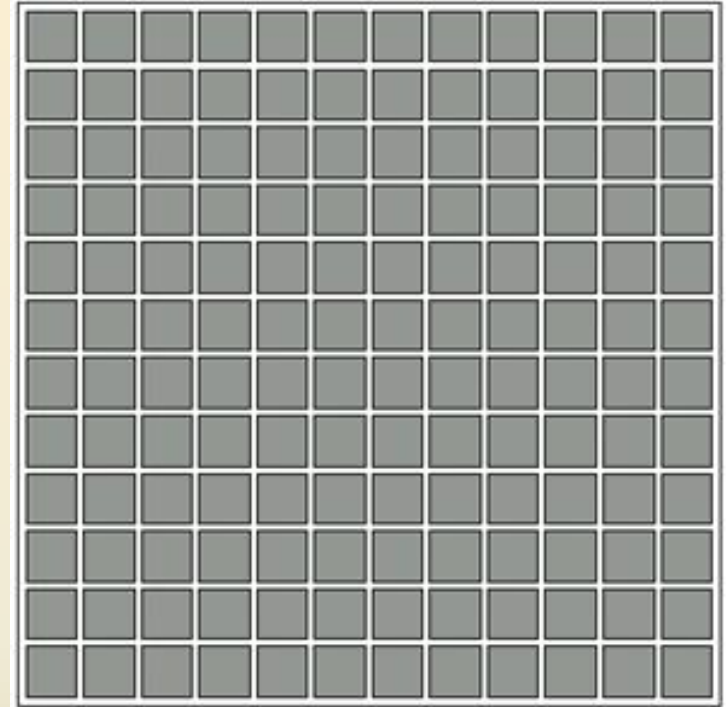
Image Sensors used to transform illumination energy to digital images.

□ 3 main sensor arrangements:

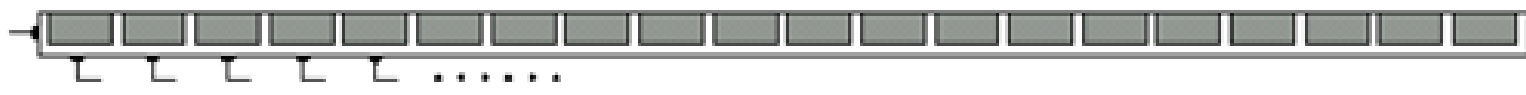
1. **Single sensor**
2. **Line sensor**
3. **Array sensor**



Single sensor



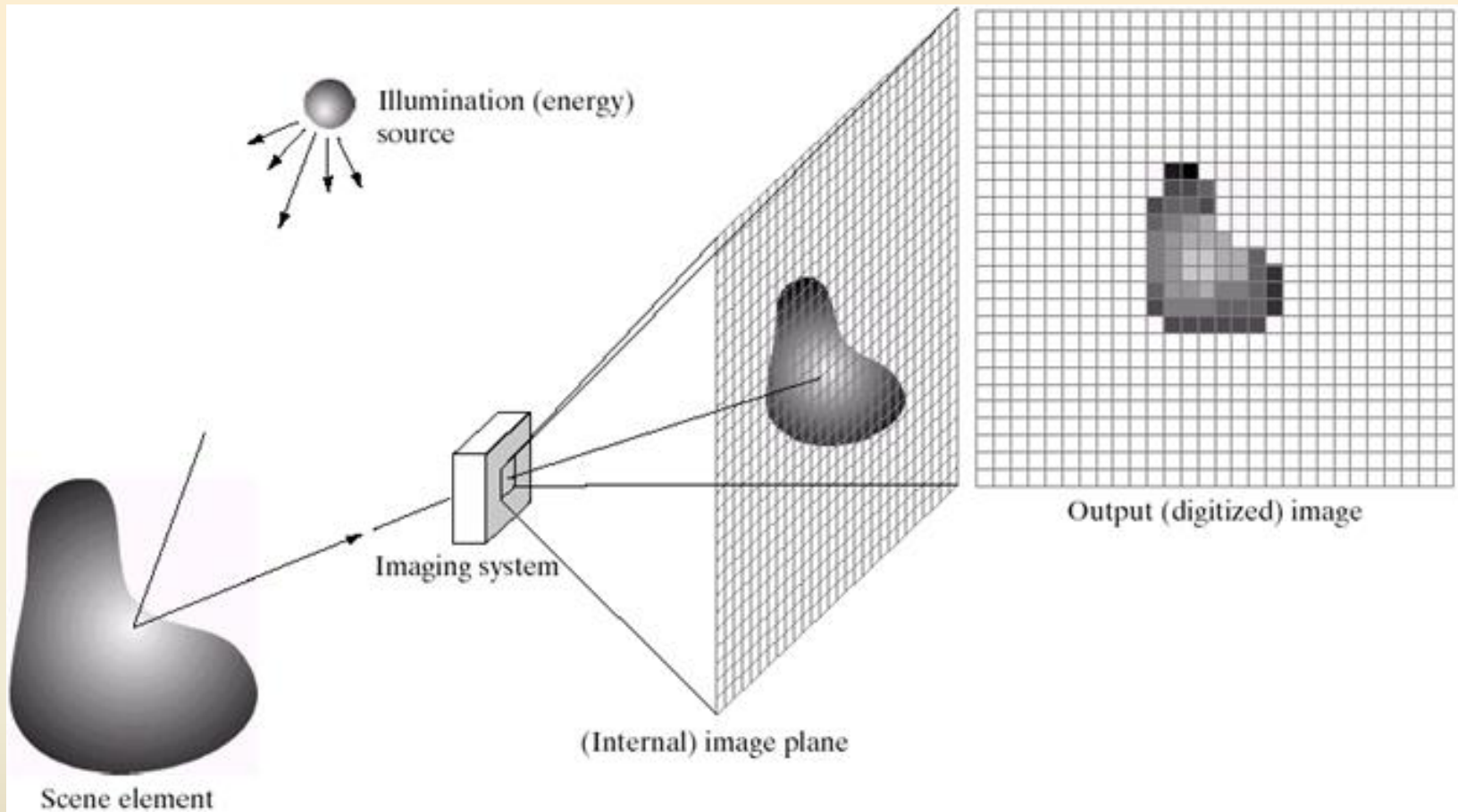
Array sensor



Line sensor

- Sensing element can be a photodiode
- Filter absorbs extra energy or acts as pass-band

Image Acquisition using Sensor Array



- ❑ Images are typically generated by *illuminating* a scene and absorbing the energy reflected by the objects in that scene.

Human Vision VS Computer Vision



What we see

0	3	2	5	4	7	6	9	8
3	0	1	2	3	4	5	6	7
2	1	0	3	2	5	4	7	6
5	2	3	0	1	2	3	4	5
4	3	2	1	0	3	2	5	4
7	4	5	2	3	0	1	2	3
6	5	4	3	2	1	0	3	2
9	6	7	4	5	2	3	0	1
8	7	6	5	4	3	2	1	0

What a computer sees

Next class >>>

- ❑ Resolutions
- ❑ Types of Digital Image
- ❑ Effect of Spatial and Gray Level Resolutions